SIMATS ENGINEERING ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I __G.Jaahnavi(192525277) _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: __G.Jaahnavi___
Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy.
How can encryption and access control protect sensitive backup data?
Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. . Cloud Auto-Scaling and Load Balancing

Answer

A company experiences unpredictable traffic spikes during seasonal sales such as Diwali, Black Friday, or New Year offers. During these periods, millions of customers visit the website at the same time. If the company uses a fixed number of servers, the servers may become overloaded, causing slow website performance, failed transactions, or even complete service failure. Cloud computing solves this problem through **Auto-Scaling**, **Load Balancing**, and **Monitoring Tools**.

Cloud Auto-Scaling

Cloud Auto-Scaling automatically adjusts the number of virtual servers based on user demand.

- During high traffic, additional server instances are automatically created.
- During low traffic, unnecessary servers are removed.
- Ensures applications always have enough computing resources.
- Prevents server overload and maintains application performance.
- Reduces operational cost because organizations pay only for the resources they use.

Load Balancing

Load Balancing distributes incoming user requests equally among multiple servers.

- Prevents one server from handling all requests.
- Improves response time and application speed.
- Provides fault tolerance by redirecting traffic if one server fails.
- Maintains continuous service availability.

Reducing Downtime and Improving User Experience

Auto-Scaling and Load Balancing together provide:

- High availability of cloud services.
- Faster webpage loading.
- Smooth checkout and payment processing.
- Reduced server crashes.
- Better customer satisfaction.
- Continuous business operations during peak demand.

Role of Monitoring Tools

Monitoring tools continuously observe cloud resources such as:

- CPU utilization
- Memory usage

- Network traffic
- Disk usage
- Application response time

When resource usage exceeds predefined limits, monitoring tools automatically trigger Auto-Scaling and notify administrators about any system issues.

Real-World Example

During **Amazon Prime Day** or **Flipkart Big Billion Days**, millions of users shop simultaneously. Cloud Auto-Scaling automatically launches new server instances to handle increased traffic. Load Balancers distribute customer requests across all servers, while monitoring tools continuously monitor system performance. This ensures uninterrupted shopping and secure payment processing.

Conclusion

Cloud Auto-Scaling, Load Balancing, and Monitoring Tools work together to improve performance, reduce downtime, optimize resource usage, and provide an excellent customer experience during heavy traffic.

2. Cloud Storage, Data Redundancy, Encryption, and Disaster Recovery

Answer

Organizations generate huge amounts of backup data that must be stored securely and reliably. Cloud storage provides scalable, durable, and cost-effective solutions for storing backup data while ensuring business continuity.

Cloud Storage

Cloud storage allows organizations to store backup data in remote cloud data centers instead of physical storage devices.

Benefits include:

- Unlimited storage capacity
- Easy accessibility
- Automatic backup
- High availability
- Low maintenance cost

Data Durability

Cloud providers maintain multiple copies of data.

- Data is replicated across several storage devices.
- Copies are stored in different geographic locations.
- If one storage device fails, another copy remains available.
- This ensures nearly 100% data durability.

Data Redundancy

Redundancy means storing duplicate copies of data.

Benefits include:

- Prevents permanent data loss.
- Supports quick recovery after hardware failures.
- Improves business continuity.
- Protects against natural disasters.

Encryption

Encryption converts readable data into unreadable coded information.

There are two types:

- Encryption at Rest – protects stored data.
- Encryption in Transit – protects data while transferring over the internet.

Only authorized users with encryption keys can access the data.

Access Control

Access control ensures that only authorized users can access backup data.

Methods include:

- Username and password
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Identity and Access Management (IAM)

Cost Advantages

Compared to traditional storage systems:

- No expensive hardware purchase.
- No maintenance costs.

- Pay-as-you-use pricing.
- Automatic software updates.
- Lower electricity and infrastructure costs.

Disaster Recovery Example

Suppose a company's office is damaged due to fire or flooding. Since backup data is stored securely in the cloud, the organization can restore all files, databases, and applications quickly without major data loss, allowing business operations to continue.

Conclusion

Cloud storage provides secure, scalable, durable, and cost-effective backup solutions with strong encryption, redundancy, and disaster recovery capabilities.

3.Content Delivery Networks (CDNs) and Global Application

1. Introduction

When a mobile application is used by people across different countries, users expect fast loading and smooth performance. If all requests are handled by one central server, users who are far away experience delays. A **Content Delivery Network (CDN)** solves this problem by delivering content from servers located close to users.

2. Content Delivery Network (CDN)

A CDN is a network of geographically distributed servers that store copies of application content.

Functions:

- Delivers content from the nearest server.
- Reduces the distance data travels.
- Improves website and application speed.
- Handles large numbers of user requests.
- Reduces the load on the main server.

3. Reducing Latency

Latency is the delay between sending a request and receiving a response.

CDNs reduce latency by:

- Delivering data from nearby edge servers.
- Reducing network congestion.
- Minimizing data travel distance.
- Improving response time.
- Providing faster access to content.

4. Role of Edge Servers

Edge servers are located close to end users.

Benefits:

- Faster content delivery.
- Reduced buffering during streaming.
- Lower response time for applications.
- Better gaming experience.
- Improved reliability and performance.

5. Global Availability

Cloud providers maintain data centers in multiple countries.

They provide:

- High availability.
- Automatic failover.
- Global load balancing.
- Disaster recovery support.
- Continuous service even if one region fails.

6. Real-Time Streaming and Gaming Example

Netflix uses CDNs to deliver movies from nearby edge servers. Users experience high-quality video with minimal buffering.

Online games like **PUBG** or **Free Fire** use edge servers to reduce network delay, resulting in faster gameplay and smoother user interactions.

7. Advantages

- Faster application performance.
- Lower latency.
- Better user experience.
- Reduced server load.
- High scalability.
- Reliable global service.

8. Conclusion

CDNs and edge servers improve application performance by reducing latency and delivering content quickly. Cloud providers ensure global availability, making CDNs essential for streaming, gaming, and other real-time applications.

4. **Cloud Security, Compliance, and Data Sovereignty**

1. Introduction

Financial institutions handle confidential customer information such as bank account details and transaction records. Cloud providers offer security features and compliance standards to protect this sensitive data and meet legal regulations.

2. Regulatory Compliance

Cloud providers support standards such as:

- ISO 27001
- PCI-DSS
- GDPR
- SOC Reports

These standards help organizations meet government and industry security requirements.

3. Auditing

Auditing is the process of reviewing system activities.

Benefits:

- Tracks user actions.

- Detects unauthorized access.
- Ensures compliance with regulations.
- Supports security investigations.
- Maintains accountability.

4. Role of Logging

Logging records all activities performed in the cloud.

It records:

- User login history.
- File access.
- Configuration changes.
- Financial transactions.
- Security events.

5. Role of Monitoring

Monitoring continuously checks cloud resources.

It helps:

- Detect suspicious activities.
- Monitor server performance.
- Identify cyberattacks.
- Generate security alerts.
- Prevent system failures.

6. Role of Encryption

Encryption converts readable information into coded data.

Types:

- Encryption at Rest.
- Encryption in Transit.

Benefits:

- Protects customer information.
- Prevents unauthorized access.
- Ensures secure communication.

- Maintains data confidentiality.

7. Data Sovereignty

Data sovereignty means storing customer data in specific countries according to legal requirements.

Organizations choose cloud regions that comply with national laws and regulations.

8. Financial Transaction Example

When customers perform online banking transactions:

- Data is encrypted.
- Every transaction is logged.
- Monitoring systems detect suspicious activities.
- Audit reports ensure regulatory compliance.
- Customer data is stored in approved cloud regions..

10. Conclusion

Cloud providers use compliance standards, auditing, logging, monitoring, encryption, and data sovereignty to provide a secure environment for financial institutions and protect sensitive customer information.

5. Containers and Kubernetes

1. Introduction

Modern applications require fast deployment, efficient resource usage, and easy scalability. Containerization packages applications with all required dependencies, making deployment simple and consistent across different cloud platforms.

2. Containers

Containers package:

- Application code.
- Required libraries.
- Runtime environment.
- Dependencies.

They share the host operating system, making them lightweight and efficient.

3. Containers vs Virtual Machines

Containers

- Lightweight.
- Fast startup.
- Low memory usage.
- Share host operating system.
- Higher application density.

Virtual Machines

- Each VM has its own operating system.
- Higher memory consumption.
- Slower startup.
- Requires more hardware resources.
- Less efficient for large-scale deployments.

4. Performance and Resource Usage

Containers:

- Start within seconds.
- Use less CPU and memory.
- Support faster deployment.
- Improve application performance.
- Reduce infrastructure costs.

5. Kubernetes Orchestration

Kubernetes automates container management.

Functions:

- Automatic deployment.
- Auto-scaling.
- Load balancing.
- Self-healing (restarts failed containers).
- Rolling updates and rollbacks.

6. Portability

Containers run consistently across:

- Private Cloud.
- Public Cloud.
- Hybrid Cloud.
- Multi-cloud environments.

This allows organizations to migrate applications easily without changing the code.

7. Microservices Example

An e-commerce application can be divided into separate containerized services:

- User Login Service.
- Product Catalog Service.
- Shopping Cart Service.
- Payment Service.
- Order Management Service.

Kubernetes manages each service independently, allowing updates and scaling without affecting the entire application.

8. Advantages

- Faster deployment.
- Better scalability.
- Lower resource usage.
- High portability.
- Simplified management.
- Improved reliability.

9. Conclusion

Containers provide lightweight, portable, and efficient application deployment, while Kubernetes automates deployment, scaling, and management. Together, they enable organizations to build reliable, scalable, and cloud-native applications.